3.

```
#include <stdio.h>

int main() {
    int choice, num, i, isPrime;

    while (1) {
        printf("\n--- Menu ---\n");
        printf("1. Check Prime\n");
        printf("2. Check Even/Odd\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        if (choice == 3) {
            printf("Exiting program.\n");
            break;
        }

        printf("Enter a number: ");
        scanf("%d", &num);

        switch (choice) {
            case 1:
                if (num < 2) {
```

```
        printf("%d is not a prime number.\n", num);
        break;
    }
    isPrime = 1;
    for (i = 2; i <= num / 2; i++) {
        if (num % i == 0) {
            isPrime = 0;
            break;
        }
    }
    if (isPrime)
        printf("%d is a prime number.\n", num);
    else
        printf("%d is not a prime number.\n", num);
    break;

case 2:
    if (num % 2 == 0)
        printf("%d is Even.\n", num);
    else
        printf("%d is Odd.\n", num);
    break;

default:
    printf("Invalid choice! Please try again.\n");
```

```
        }
    }

    return 0;
}
```

OUTPUT:

```
1. Check Prime
2. Check Even/Odd
3. Exit
Enter your choice: 1
Enter a number: 7
7 is a prime number.

--- Menu ---
1. Check Prime
2. Check Even/Odd
3. Exit
Enter your choice: 2
Enter a number: 87
87 is Odd.
```